Jay Kamavisdar

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About

Undergraduate (Pre Final Year Student) in Computer Science Engineering (AI/ML) with a strong foundation in machine learning, probability, statistics, and data structures. Proficient in Java and Python, with hands-on experience using Jupyter, Miniconda, and Git for experimentation and project development. Worked on practical ML/DL projects involving CNNs, model evaluation, and experimental analysis. Familiar with AWS services including EC2, S3, IAM, Lambda, and CloudWatch. Interested in applied machine learning, model optimization, and solving real-world problems using data-driven approaches.

Publications

Fine-Tuning for Code Intelligence: Evaluating LLMs on Custom Programming Benchmarks — July 2025

Harsh Sahu, Munsifa Firdaus Khan Barbhuyan, ***Jay Kamavisdar***, Ayushman Mishra, Rishabh Pandey, Samarth Pratap Singh

Status: Accepted and presented at an international conference; proceedings under publication (2025)

Education

Kendriya Vidyalaya Masjid Moth — April 2020 - May 2023

- **Class X th (CBSE)**: 94.4% (Marksheet here)
- **Class XII th (CBSE)**: 89.2% (Best of 5 with Core Subjects+Language) (Marksheet here)

VIT Bhopal University, B.Tech. in Computer Science Engineering (AI/ML Specialization) — Sept 2023 – May 2027

- **CGPA**: 8.71/10.0 (Till 6th Semester)

Projects

Image Restoration under Synthetic Degradation using Deep Learning — Kaggle Here

- Designed a controlled degradation pipeline simulating noise, blur, compression artifacts, and resolution loss to generate paired training data.
- Trained and evaluated deep learning models for image reconstruction and denoising, optimizing loss functions and training schedules for stability and generalization.
- Assessed restoration quality using quantitative metrics and qualitative visual analysis on unseen degraded samples.
- *Tech:* Python, PyTorch, OpenCV, NumPy

Oceanographic Data Retrieval and Analysis using ARGO Floats — Kaggle Here

- Accessed and processed large-scale oceanographic datasets using argopy framework and ERDDAP services, working with multi-diimensional scientific data.
- Performed regional and depth-wise analysis of temperature profiles using xarray-based workflows to validate data integrity and exploratory needs
- *Tech:* Python, Argopy, xarray, NumPy, ERDDAP

GarudaVPN — GitHub Here

- Designing a secure VPN architecture using modern cryptographic primitives (X25519 key exchange, HKDF-based key derivation, AES encryption) for authenticated and confidential communication.
- Implementing secure socket communication and integrating OpenVPN concepts; evaluating protocol correctness and security trade-offs.
- *Tech:* Java, OpenVPN, BouncyCastle, Java Cryptography APIs

Water Quality Analysis using Machine Learning GitHub Here

- Developed an end-to-end ML pipeline including data preprocessing, feature selection, model training, and evaluation on real-world environmental data.
- Analyzed model performance using quantitative metrics and visualized insights through interactive dashboards.
- *Tech:* Python, scikit-learn, NumPy, Matplotlib, Power BI

Technologies

Languages: Python, Java, C++
Machine Learning: PyTorch, scikit-learn
Data & Visualization: Numpy, Pandas, Matplotlib, Power BI
Systems & Tools: Git, Jupyter
Security: Java Cryptography, BouncyCastle
Databases: MySQL, MongoDB

Certifications

Cloud Computing May 2025

- Issuing Authority - NPTEL, India (Certificate)
- **Coursework:** Virtualization, Cloud Computing

Applied Data Analytics June 2025

- Issuing Authority - CEC, NIT Raipur (Certificate)
- **Coursework:** Data Analytics, Power BI, Matplotlib